

RC2 Temporal Sensor Protocol

Field Equipment

Sensors

- **Ultrameter** (pH, SpC, and TDS)
- **Spectrolyzer** (NO₃, turbidity, DOC) – probe, 10m extension cable, and equipment in orange Pelican case, which includes
 - Screen
 - 12V battery
 - Battery connectors and fuses (including extra cables and fuses) battery connectors and fuses (case)
- **Manta** (Turbidity, SpC, temperature, and depth)
 - Weighted sensor guard and storage cup
 - Manta pipe kit with rope and carabiner
- **Rugged baroTROLL** + rope to hang the sensor (barometric pressure)
- **YSI** (DO mg/L, DO%, mmHg)
- 2 kneeling pads
- Data sheet holder with metadata sheets, pencils, pens, sharpies
- 2 backpacks
- Tool bag (at a minimum, there should be a Manta screwdriver in one of the backpacks to tighten the eye hook after logging is turned on)

Sensor Calibrations

Calibrate sensors a day prior to going out to the field

Manta

To calibrate a sensor for the first time, check the calibration, or recalibrate, follow the steps outlined below:

1. Remove the calibration cup from sensor body and set the cup in the sink. Gently set the sensor body on the sink counter. **NOTE: Be sure to brace the sensor body with the weighted sensor guard or another heavy object when you set it on the counter so it won't roll off the counter and onto the floor.**
2. Rinse the sensor heads and calibration cup with tap water under the sink at least three times to remove any buildup of salts and/or dried sediments. Make sure that the sensor heads are clean and in good condition, i.e., no sediment or algal film build up or scratches on the sensor face(s).
3. To prevent dilution of the calibration standards that may result from remaining water droplets in the cup and on the sensor bodies, rinse the sensor heads and calibration cup with the appropriate standard (see above). This is done by filling the calibration cup with ¼" - ½" of the appropriate standard and then hand-tighten the sensor body onto the calibration cup. Gently swish the solution around in the cup to rinse all exposed surfaces by gently turning the sensor upside down and then right-side-up at least 5-10 times. Then hold the Manta horizontal and rotate the sensor 360° at least three times. Unscrew the cup and empty the calibration solution either into the sink (pH and turbidity standard) or in the waste container (conductivity standard).

4. Place the sensor body—with the sensor heads facing up—in a ring stand with clamps. Place a heavy object (a second Manta with a weighted sensor guard works well) on the base of the ring stand to counter-balance the clamped Manta). Tighten the clamps around the body as much as you can so the body doesn't slip. It helps to place the clamps just below the screws connecting the housing to the sensor body, with each arm below a screw (the arms “catch” on the screws and help hold the body in place). Be sure to set the body high enough in the ring stand above the bench to allow enough room to fit the Data Cable, allowing the cable to bend gently. If you can, tilt the sensor in the ring stand to minimize creating bubbles when you pour the standard in the cup. Slowly fill the calibration cup with the appropriate standard so that there is a minimum of 1” above the head of the tallest sensor. Lightly tap the cup to make sure there aren't any air bubbles trapped in the sensor faces, including the conductivity sensor slot.
5. Connect the Manta to your computer using the Data Cable and USB Converter and open the Manta 2 Control Software program. Allow 5 minutes or so for the sensor to equilibrate with the calibration solution. Check the sensor reading of the calibration standard.
 - a. If the value is $\leq \pm 10\%$ of the calibration standard value, **DO NOT calibrate the sensor.**
 - b. If the value is **greater than $\pm 10\%$ of the standard value, go to Step #6 and calibrate the sensor.**
6. Click the **Manta2** menu, move your mouse down to the “Calibrate...” option, slide your mouse to the right to the open the parameter selection box, and select “SpCond $\mu\text{S}/\text{cm}$ ” (fourth line down) to open the “Calibrate Sensor SpCond $\mu\text{S}/\text{cm}$ ” pop-up box (Figure 1). This box will walk you through the calibration.

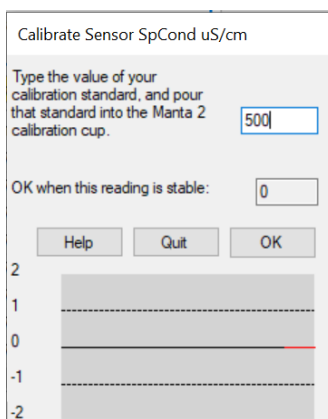


Figure 1. Calibrate menu pop-up box to “Calibrate Sensor SpCond $\mu\text{S}/\text{cm}$.”

7. Type the calibration standard value (e.g., “500” for conductivity, “100” for turbidity, and “7” for pH) in the blank box to enter the value of the calibration standard and wait for the value to stabilize. Click “OK” when the sensor reading is stable, i.e., the value in the text box and on the graph fluctuates within ± 5 units (see Figure 1).
8. The software will report back a “Sensor Response Factor” (SRF), which is automatically generated by the Manta software (Figure 2). The SRF value reflects how close the sensor reading of the calibration standard is to the actual calibration standard value, which in this case is 500 $\mu\text{S}/\text{cm}$. If the recalibrated sensor value equals or is within ± 2 units of the target calibration standard value, and the SRF is within the acceptable range stated above in Step 8 (above), click “OK” to accept the calibration. If not, try the calibration again until the value equals the

standard value and the sensor returns an acceptable SRF. As stated by Eureka in the User's Manual (p. 62):

"Suppose that a typical Conductivity sensor reports 100 μ A in a 1413 μ S/cm standard. If your particular Conductivity sensor reports 100 μ A in that same calibration solution, then your SRF is 100% (some parameters, such as pH, have a more complex SRF calculation, but the effect is the same). If your response is 80 μ A, your SRF would be 80%. When you click the OK button to accept a calibration, the Manta automatically accepts your calibration if the SRF is between 60% and 140%. If the SRF falls outside that range, you will be cautioned to check your standard value, make sure the sensor is clean, make sure the reading has stabilized, etc. But you can elect to accept any SRF. Each sensor calibration's Sensor Response Factor (SRF) is automatically logged into the Cal Record (i.e., cal.log file) with the details of that calibration."

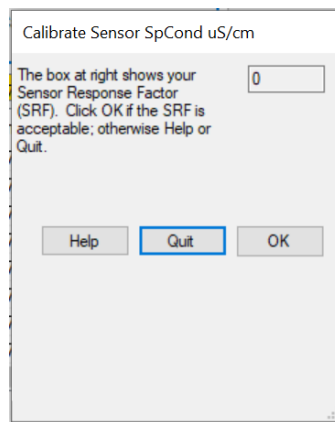


Figure 2. "Calibrate Sensor SpCond μ S/cm Sensor Response Factor (SRF) pop-up box.

- Calibration data is stored in the cal.log file. To view the cal.log file, click the **Manta2** menu and select the "Cal Log..." option, which brings up a "Viewing" box with the calibration data for every calibration performed on that sensor (Figure 3). To download the cal.log file, click "Save As..." in the cal.log "Viewing" box and save the file as a .csv file to the RC2 Temporal Study ShareDrive

Date	Time	Sensor	SN	Units	RV	Old	New	SRF	
03/18/21	10:38:38	DPTH	07151546	m	1.452121e+...	10.22	0.00	98	Done
03/18/21	10:39:17	TURB	03214660	FNU	2.113740e-02	0.01	0.00	N/A	Done
03/18/21	10:39:53	TURB	03214660	FNU	9.461290e-01	618.0	600.0	81	Done
03/18/21	10:40:41	COND	03214660	μ S/cm	2.914962e+...	1412	1413	99	Done
08/27/21	17:22:00	COND	03214660	μ S/cm	7.755509e+...	529.6	500.0	105	Done
08/27/21	17:58:12	TURB	03214660	FNU	1.284730e-01	67.85	100.0	113	Done
09/13/21	13:29:44	COND	03214660	μ S/cm	7.933150e+...	495.8	500.0	104	Done
09/22/21	09:48:42	COND	03214660	μ S/cm	7.939662e+...	502.4	500.0	105	Done
09/22/21	09:58:16	TURB	03214660	FNU	1.312760e-01	102.8	100.0	110	Done

Figure 3. Pop-up box for viewing the calibration log file ("cal.log").

(see Appendix for link).

10. Calibration results are recorded in the Excel spreadsheet titled "RC2_Calibration_Log.xls," which is stored on the RC2 Teams channel "Sensor Data and Metadata" (see Appendix for link). When you are satisfied with the calibration, record the following values in the appropriate column for each parameter calibrated:
 - a. Date: date calibration or calibration check was performed.
 - b. Battery_V: remaining internal battery voltage (to measure this value, you must turn the battery switch to "ON" to activate battery power).
 - c. XXX_target_YYY, where "XXX" = the associated target calibration standard (abbreviated) and "YYY" = the associated standard units. **(NOTE: Make sure the units are correct!)**
 - d. XXX_final_YYY, where "XXX" = the final measured sensor value of the associated calibration standard (whether calibrated or not) and "YYY" = the associated standard units.
 - e. XXX_calibrated: "YES" or "NO" indicating whether the sensor was or was not calibrated
 - i. Record "NO": sensor readings were within the acceptable calibration thresholds ($\pm 10\%$ of the standard value) and the sensor was not calibrated
 - ii. Record "YES": sensor readings were outside the acceptable calibration thresholds ($>10\%$ of the standard value) and the sensor was calibrated

YSI EcoSense ODO200M

- Calibrate following the "DO% in water saturated air 1 point calibration" method outlined in the [manual](#)

Ultrameter

- Calibrate each sensor following the [manual](#):
 - pH
 - SpC
 - TDS

Spectrolyzer

1. Do not calibrate but take a blank following these steps:
 - a. Rinse the probe and white sleeve cup with milli-Q water 3 times.
 - b. Using 70% ethanol and a cotton swab, carefully clean the optical windows on the probe.
 - c. Attach the white sleeve cup over the optical sensing portion of the probe, fill the sleeve cup with milli-Q water, and attach to the screen.
 - d. Connect the battery to the screen
 - e. On the screen, on the left side go to **Service**
 - i. Then **Terminal-->Settings-->Station --> Name the station by adding "Blank_Date" --> Save**
 - f. Click on Service again
 - i. Then **Measurement --> Terminal --> Measurement --> Automatic --> Set measurement interval to 102 --> Save**
 - g. Click on **Service** once more and make sure to hit "**Leave Service Mode**"
 - h. Check the lower right-hand corner for the green "**Measuring**" status
 - i. Record the start time using the clock on the screen
 - j. Log for at least 8.5 minutes, record end time, then download

Deploying the sensors

Rugged BaroTROLL

1. When you first arrive at the site, locate a tree or bush where you can tie the barotroll to.
2. Do not place on the ground
3. The location needs to be representative of where you are sampling.
4. Record the time you set the sensor in place

Note: This sensor DOES NOT go in the water

5. Leave the barotroll up during the during the entire duration of the sampling.
6. Retrieve the barotroll once water sampling has been completed (~20 minutes from arriving at the site).
7. Record the time you retrieved the barotroll

Spectrolyzer

1. Place the probe into the river and make sure it is fully submerged with the optical sensing area facing up. Do not place in anything that will obstruct the optical sensing window.
2. On the screen, on the left side go to **Service**
 - a. Then **Terminal --> Settings --> Station --> Name the station with "River_Date" --> save**
3. Click on **Service** again
 - a. **Measurement --> Terminal --> Measurement --> Automatic --> Set measurement interval to 102 --> Save**
4. Click on **Service** once more and make sure to hit **"Leave Service Mode"**
5. Check the lower right-hand corner for the green **"Measuring"** status
6. Record the start time using the clock on the scan screen
7. Log for at least 8.5 minutes, record end time, then download
 - a. Insert your thumb drive to the USB port
 - b. Go to **Service --> Outputs --> File Output --> Manual Download --> Fingerprint**
 - i. Change the "From:" to the date and time you started logging. Make sure to choose the correct month and day. And input the time in PDT
 - ii. The "To:" you don't need to change because it will automatically fill in the time you are downloading.
 - c. Click **download --> Save As --> USB --> Navigate to location on USB --> save**
8. Disconnect the power and disconnect the probe.

Manta

1. When you arrive at the first site of the day—before you stow all the sampling equipment in the backpack—remove the Manta storage/calibration cup and replace it with the weighted sensor guard. Dump the excess water in the cup and stow the cup in the vehicle.
2. Turn logging on
 - a. Turn the Manta battery switch to "ON". The battery switch connects ("ON") and disconnects ("OFF") the power (i.e., batteries) to the sensor's circuitry, effectively turning the sensor on and off.

- b. Remember to look for the red LED to blink five times to confirm that Logging is activated, and the green LED blink briefly to confirm that the Manta is receiving adequate voltage to start Logging. The Manta is now logging and will continue logging until you turn the battery switch to its “OFF” position, or your batteries are depleted.
3. Gently slip the Manta into the pipe kit by holding the probe by the eye bolt and lowering the Manta into the pipe. Make sure the Manta is seated on the mesh bottom of the pipe. Be careful when lowering the probe as the sensor guard can get hung up inside the pipe on the lip of the mesh cap near the bottom, and then can drop suddenly to the bottom. The Manta is built tough, but nonetheless should be handled with care.
4. Close and secure the lid using the carabiner. Be sure to run the carabiner through the loop of rope connected to the eye bolt on the lid (Figure 4).



Figure 4. Secure the pipe kit lid using the carabiner, making sure the rope loop through the carabiner.

5. **Record the “Time turned on (in the morning @ 1st site)” on the (meta)data sheet. USE YOUR FIELD WATCHES TO ENSURE YOU ARE RECORDING ALL TIMES IN PACIFIC STANDARD TIME (PST). BE SURE TO SYNCHRONIZE YOUR WATCHES AT THE START OF EACH DAY.**
6. Find a representative section of the river to deploy the sensor, and gently lay the pipe kit/Manta on the riverbed as close to the thalweg as you feel comfortable wading but without taking more than a few “incidental” steps on the riverbed. Wherever possible, do not deploy the Manta in or immediately below a riffle or in a deep pool, unless a pool is the only section of river that can accommodate the probe.
7. Tell your partner once the Manta is placed below the surface of the river (i.e., the sensors is “in the water”) and have them **record the “Time in water (at site)” on the (meta)data sheet.**
8. Secure the pipe kit rope to either a large rock or a tree, whichever is more convenient (and secure, if stream discharge is high/fast).
9. If the river is too shallow to accommodate the sensor AND the pipe kit, you can remove the Manta and simply lay the Manta on the riverbed. If necessary, attach a rope to the Manta’s eye bolt and secure the rope to the shore (see #6, above).
10. Once all other site data has been collected (~20 minutes from arrival to the site), retrieve the Manta from the river, and **record the time the sensor was pulled out of the water as the “Time**

out of water (at site)” on the (meta)data sheet. **DO NOT TOUCH THE BATTERY SWITCH UNLESS IT IS THE LAST SITE OF THE DAY (see next section).**

11. At the last site of the day

- a. Retrieve the Manta from the river at the last site of the day and record the “Time out of water (at site)” on the (meta)data sheet (see #5, previous section).
- b. Pull the Manta out of the pipe kit and use a rag to dry the top of the Manta, particularly around the battery plug to prevent water from seeping into the IBP (which can corrode the batteries or cause the system to short circuit).
- c. Loosen the steel eye bolt on the battery plug until you can swing the battery switch back and forth.
- d. Move the switch to “OFF”—disconnecting the power supplied to the sensor—and hand-tighten the eye bolt. The battery plug does not need to be overtightened when simply turning the switch off. **NOTE: Be careful to firmly seat the small round peg underneath the switch in the round notch before you tighten the eye bolt.** If the switch is not seated correctly, the switch can snap in two as you tighten the eye bolt.
- e. Record the time the battery switch is turned to “OFF” on the (meta)sheet as the “Time turned off (in the afternoon @ last site)”.

YSI

1. Turn the YSI on and take off the grey sleeve.
2. Place in the stream in a similar location to the Manta, making sure there is no vegetation blocking the probe.
3. Let equilibrate for 5 minutes.
4. Take 2 readings throughout the duration of sampling. Once in the beginning and once near the end.
5. Record DO%, DO mg/L, and temperature in deg C in the metadata sheet. Use the “Mode” button to toggle between those settings.

Ultrameter

1. Unplug the pH port and dip the cup portion of the ultrameter into the stream.
2. Repeat this 2 more times before taking the actual reading.
3. Press “Cond” for SpC, “pH” for pH, and “TDS” for TDS
4. Wait a few seconds for the reading to stabilize.
5. Record the reading along with the temperature. (Note: The reading disappears about 10 seconds after you press the keys)
6. Take 3 consecutive readings for each of the analytes and record on the metadata sheet along with the time and temperature.